# AYKUT DAĞ

**Architectural Designer | Computational Designer** Parametric Design • Generative AI • Digital Fabrication

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## PROFESSIONAL SUMMARY

Architectural and computational designer with 4+ years of experience in international design competitions and concept design for cultural, hospitality and office buildings. Works across the whole concept stage, from massing, space programming and design development to visualization and presentation, and has led multidisciplinary concept teams through competition submissions, including a shortlisted finalist entry. Specializes in parametric design and algorithmic modeling in Rhino and Grasshopper, generating design options and rationalizing complex geometry for fabrication, and uses AI coding agents to build Python scripts and small automation tools. Extends this practice into interactive and responsive environments with TouchDesigner and sensor data, generative AI for concept exploration, and hands-on digital fabrication and prototyping, with a particular interest in biophilic and bio-inspired design. Currently pursuing an M.Sc. in Architectural Design Computing at Istanbul Technical University, researching pneumatic systems and soft robotics. Comfortable working in international, remote and multicultural teams.

**Core tools:** Rhinoceros 3D, Grasshopper, TouchDesigner, AutoCAD, Enscape, V-Ray, Lumion, Twinmotion, ComfyUI, Adobe Creative Cloud

## PROFESSIONAL EXPERIENCE

**GG-loop** — Architectural Designer (Facade Design & Computational Visualization) *2024 – 2025*

*Amsterdam, Netherlands (Remote collaboration)*

Worked with the Amsterdam-based biophilic architecture studio on facade design, parametric modeling and visualization at concept stage.

- Developed parametric facade and envelope geometry in Rhino and Grasshopper, from concept options to detailed balcony and shading elements.
- Produced photorealistic renders and concept presentation decks for client reviews.
- Set up the digital modeling pipeline from early concept sketches to refined geometry and presentation visuals.

**Selected project: Viridia Hotel**, Kavaja, Albania (114,000 m², 140 m high-rise resort tower, with FOCUS Architecture): biophilic facade concept and parametric balcony geometry.

**Melike Altinisik Architects (MAA)** — Concept Team Leader & Architect *2021 – 2024*

*Istanbul, Turkey – Intern (2021) → Architect (2021–2023) → Concept Team Leader (2023–2024)*

Concept design and international competition work on cultural, hospitality and office buildings, from early massing to final submission.

- Led concept teams from early design to final submission: massing, space programming, area schedules, 2D drawings and jury presentations.
- Built custom Grasshopper definitions for double-curved facade panelization, parametric form-finding and geometric rationalization.
- Managed the production of competition boards, project books and brochures; supervised and mentored interns.
- Introduced Stable Diffusion and ComfyUI into concept ideation, used Enscape/Lumion VR in design reviews, and ran the office 3D-printing workflow (Ultimaker, Creality) for scale models.

**Selected projects: II. Sejong Cultural Center for Performing Arts**, Seoul (57,645 m², with BAUM Architects; shortlisted finalist, competition team leader) · **Libreville Hotel & Congress Center**, Gabon · **Turkish Petroleum Headquarters**, Istanbul · **Solar Pavilion** (product design) · **Incheon Geomdan Museum Library Cultural Complex**, Korea

**Aboutblank** — Architectural Intern, Competition Team *2020 – 2021*

*Istanbul, Turkey*

Competition team member on urban design and public-space competitions: spatial diagrams, urban mobility mapping and 2D drawings.

**Selected project: Kadıköy Square Urban Design Competition**, Istanbul (1st Prize, with Adapt Architects)

**Nous Architecture** — Competition Team Member *2020*

*Istanbul, Turkey*

Competition work on waterfront public space: space typologies, environmental sections and urban connectivity diagrams.

**Selected project: Haliç Seashore Design Competition**, Istanbul (3rd Prize)

## COMPETITIONS & HONORS

- **II. Sejong Cultural Center for Performing Arts:** Shortlisted Finalist, International Competition (Seoul, South Korea, 2023), Concept Team Leader at MAA
- **Kadıköy Square Urban Design Competition:** 1st Prize (Istanbul, 2020), with Aboutblank & Adapt Architects
- **Haliç Seashore Design Competition:** 3rd Prize (Istanbul, 2020), with Nous Architecture
- **1927 Mimarlar Derneği:** 3rd Prize, Architectural Drawing & Photography Competition (2020)

## TECHNICAL SKILLS

**Computational Design:** Rhinoceros 3D, Grasshopper, TouchDesigner, Parametric Modeling, Algorithmic Design, Geometric Rationalization, Facade Panelization.

**Architectural Design & Documentation:** Concept Design, Competition Design, Massing & Space Programming, AutoCAD, Revit, Autodesk Forma.

**Visualization:** Enscape (VR), V-Ray, Lumion, Twinmotion, D5 Render, Adobe Creative Cloud (Photoshop, InDesign, Illustrator).

**AI-Assisted Development:** AI coding agents (Claude Code, Google Antigravity) for Python scripts, Grasshopper components and workflow automation.

**Generative AI & Interactive Media:** Stable Diffusion, ComfyUI, Runway ML, StyleGAN, Real-Time Interactive Systems, Arduino Sensors, EEG/Biosignal-Driven Projection Mapping.

**Digital Fabrication:** 3D Printing (Ultimaker, Creality, Cura), Soft Robotic Actuators, Silicone Casting (Smooth-On Ecoflex), Fabric Formwork, Physical Prototyping.

## EDUCATION

**Istanbul Technical University (İTÜ)** — **M.Sc. in Architectural Design Computing** *2023 – Present*
Department of Informatics | Focus: Soft Robotics, Bio-Inspired Interactive Systems, Pneumatics, Material-Driven Computation

**Bauhaus-Universität Weimar** — **Erasmus+ Exchange, M.Arch. Program** *2024 – 2025*
Faculty of Architecture and Urbanism, Weimar, Germany

**Istanbul Technical University (İTÜ)** — **Bachelor of Architecture (B.Arch.)** *2016 – 2021*
Faculty of Architecture | Erasmus Exchange: Opole University of Technology, Poland (2018–2019)

## RESEARCH & PUBLICATIONS

**M.Sc. Thesis: "Immaterial Architectural Space: A Proposal for a Pneumatic Modular System" (2025 – Present)**
*Advisor: Assoc. Prof. Dr. Sevil Yazıcı.* A bio-inspired pneumatic modular system developed through a Research through Design (RtD) framework, translating alveolar structures and Murray's Law into responsive spatial enclosures. Built a multi-material casting workflow combining platinum-cure silicone (Smooth-On Ecoflex) with 3D-printed TPU constraint networks cast in FDM-printed negative molds, testing translucency and tactile behavior under air pressure.

**Publication: "Pulse Pavilion: A Human-Driven Psychophysiological Spatial Feedback System"**
Co-author with KHORA Collective (Uçkan, Atukeren, Dağ et al.). Abstract accepted for the Digital Futures 2026 conference.

**Biomimetic Machine Learning (StyleGAN)**
Trained StyleGAN models to blend historic urban morphology with the growth patterns of *Physarum polycephalum* (slime mold) as a generator for emergent urban layouts.

**Deployable Fabric Formwork ("Concrete Flowers")**
Designed a deployable mold system combining parametric modeling, 3D-printed articulated frames and flexible fabric membranes to cast variable concrete panels.

**Interactive Sensor-Driven Environment ("The Orchestra of Illumination")**
Built an interactive installation that maps touch and pressure, read by piezoelectric sensors, to real-time light and sound responses.

## EXHIBITIONS & CULTURAL PROJECTS

- **[KHORA]: Space of Becoming with Mind & Body:** Spatial & Generative Designer, KHORA Collective. Istanbul Digital Art Festival 2026 (IDAF), Atatürk Kültür Merkezi (AKM), June 2026. A responsive environment that turns real-time EEG and heart-rate data into projection mapping and spatial audio (TouchDesigner).
- **Archithese Magazine, Zurich (2020):** Author & artist of the text and computational artwork "Glitch | Queer | Space" (Issue: Queer, pp. 68–69).
- **Doğuş Group "Sanata Bi Yer" Exhibition (Istanbul, 2019):** Exhibited architectural photography.

## LANGUAGES

**English:** C1 (Professional Working Proficiency) | **Turkish:** Native

## REFERENCES

**Melike Altınışık**
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